Case Competition Checkpoint 2 - Analytical Coding Implementation

Start Assignment

- Due Wednesday by 11:59pm
- Points 30
- Submitting a website url

Purpose
In this checkpoint, you will implement your Checkpoint 1 analytical plan using Python in Google Colab.
Your goal is to:

- Load, clean, and transform the City of Norman datasets
- Compute the case metrics (V/C ratio, collision counts, and rankings) and address your chosen business question (one of the City's questions Q1–Q5)

Note: Visualization, including the ward map and heat map, will be covered in Checkpoint 3.

Deliverables

1. Set up a shared Google Drive folder

- The group leader creates a shared folder named `MIS3213_FS26_Group10X` and shares it with:
  - All group members
  - Instructor: **tracychaw@gmail.com (mailto:tracychaw@gmail.com)**
  - TA Rylan: **rylant910@gmail.com (mailto:rylant910@gmail.com)**
  - TA Joshua: **joshuabarronporras@gmail.com (mailto:joshuabarronporras@gmail.com)**
  - TA Angelina: **Angelinalang125@gmail.com (mailto:Angelinalang125@gmail.com)**

2. Conduct the data analysis

Your notebook must include the following sections, each starting with a markdown heading:

- **Problem Restatement (5 pts):** Restate your chosen business question (Q1–Q5) and say how your analysis will answer it.
- **Data Loading & Inspection (5 pts):** Load the five ACOG count files (FY19, FY21, FY23, FY25, FY26), the collision workbook (`Base_Data` sheet), the `Code_Book`, and the ward GeoJSON. Show the shape, columns, and the data problems you notice in each.
- **Data Preparation & Transformation (5 pts):** Clean and combine the data following the City's clarifications.
  - Where a location appears in more than one count or year, use the highest volume.
  - Where the `Latest` column is missing, sum the directional counts: column L in FY23 and FY25, column J in FY26. FY19 has a second count in column N.
  - Define road segments of at least one mile.
  - Place each collision on the road where it occurred, using its direction and distance from the intersection.
- **Metric Computation (5 pts):** Compute the V/C ratio for each segment, using the ACOG capacity table and your lane counts. Count collisions per segment, and build the rankings your question needs.
- **Business Question (5 pts):** Answer your chosen question with ranked lists. Briefly explain the key choices you made and why.
- **File naming (5 pts):** Rename your notebook `Group###_LastName_FirstName.ipynb` and upload it to the shared folder.

3. Use the Norman Case Tutor (optional, recommended)

The **Norman Case Tutor (https://claude.ai/artifact/8XkSeSavnmcnrNVBw5oFxP)** is an AI tutor built for this case. It can explain the City's questions, help you debug your code, and check your results.

- To give it your notebook:
  1. In Colab, run all cells.
  2. Download the notebook (**File › Download › Download .ipynb**).
  3. Drag the file into the tutor.
- The tutor coaches you. It will not write your solution. Your code and your explanations must be your own work.
- At the end of your notebook, add a short markdown cell titled **AI Tutor Use**. List two or three questions you asked the tutor and what you changed as a result. Write "None" if you didn't use it.

4. Share and submit

- Click **Share** at the top right of your Colab notebook.
- Set sharing permissions to **Anyone with the link can edit**.
- Copy the shared link.
- Each member submits their own notebook link on the Canvas assignment page.